BOX 16.2 Guidelines for Stabilization Training: Principles and Progression

1. Begin training awareness of safe spinal motions and the neutral spine position or bias.
2. Have patient learn to activate the deep (core) stabilizing musculature while in the neutral position.
3. Add extremity motions to load the global musculature while maintaining a stable neutral spine position (dynamic stabilization).
4. Increase repetitions to improve holding capacity (endurance) in the stabilizing musculature; increase load (change lever arm or add resistance) to improve strength while maintaining a stable neutral spine position.
5. Use alternating isometric contractions and rhythmic stabilization techniques to enhance stabilization and balance with fluctuating loads.
6. Progress to movement from one position to another in conjunction with extremity motions while maintaining a stable neutral spine (transitional stabilization).
7. Use unstable surfaces to improve the stabilizing response and improve balance.

prohibitively expensive to use clinically for training activation of the core musculature. As an alternative device, a pressure biofeedback unit (Stabilizer™; © 2006 Encore Medical, L.P.) has been developed and has been shown to have clinical usage in training activation and control of the core musculature of the trunk and neck.[22,49]

Cervical Musculature

In the cervical region, the goal is to activate and control the muscles that control axial extension (cervical retraction). This requires capital flexion, slight flattening of the cervical lordosis, and flattening of the upper thoracic kyphosis (Fig. 16.17).

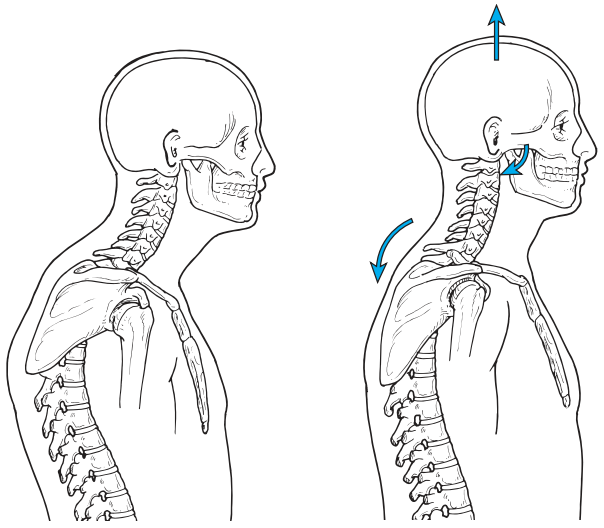


FIGURE 16.17 Axial extension (cervical retraction) involves the motions of capital flexion and extension of the lower and upper thoracic spine toward extension, resulting in slight flattening of the cervical lordosis.

Deep Neck Flexors: Activation and Training

Patient position and procedure: Supine. For craniocervical flexion and gentle axial extension, teach the patient to perform slow, controlled nodding motion of the head on the upper cervical spine ("yes" motion). Once able to activate the motion, the Stabilizer™ may be used to monitor the amount of cervical flattening and measure the muscular endurance for holding the contraction (Fig. 16.18).

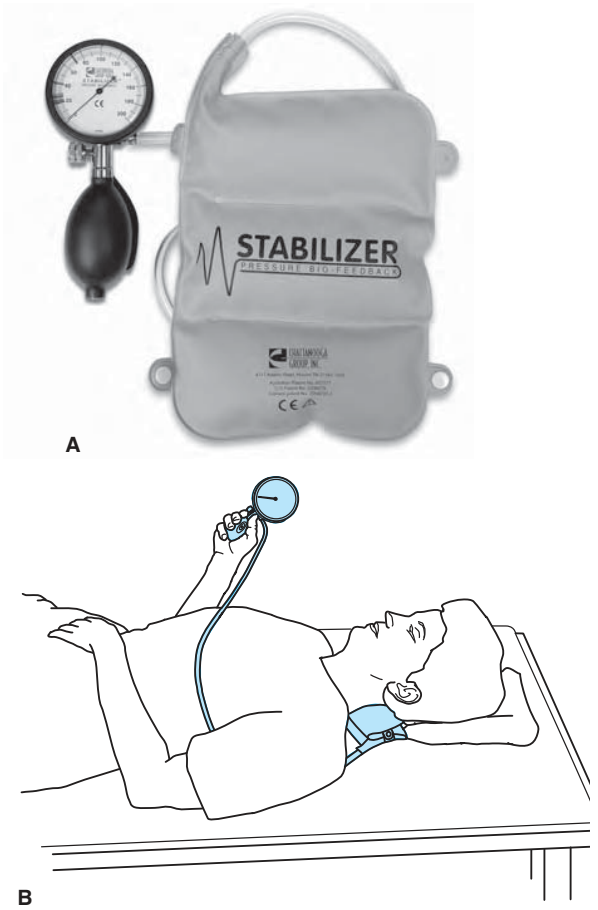


FIGURE 16.18 (*A*) The Stabilizer™ pressure biofeedback unit (© 2006 Encore Medical, L.P.) is used to provide visual feedback to the patient while training for core muscle activation. (*B*) Stabilizer folded into thirds under the cervical spine to test and train capital flexion with neutral spine axial extension.

The protocol for use of the Stabilizer™ is summarized in Box 16.3.

Focus on Evidence

Jull et al.[25] reported that the controlled performance of upper cervical flexion increases the pressure in the Stabilizer™ to 30 mmHg and that the test–retest reliability of the craniocervical flexion test (conducted on 50 asymptomatic subjects 1 week between tests) was an ICC of 0.81 for the activation score and 0.93 for the performance index (see Box 16.3).

BOX 16.3 Testing and Training Core Muscle Activation in the Cervical Spine

- Place the folded Stabilizer™ pressure biofeedback unit (folded into thirds) under the upper cervical spine and inflate to 20 mmHg.
- Instruct the patient to nod and increase pressure on the cuff to 22 mmHg and hold the pressure steady.
- If the patient is successful (i.e., can hold the position with minimal superficial muscle activity), have him or her relax and repeat the flexion, this time increasing pressure to 24 mmHg. Repeat this incremental activation up to 30 mmHg (total 10 mmHg increase).
- The final pressure is the one at which the patient can hold steady.
- Muscle endurance (holding or tonic capacity) of the deep neck flexors is measured by the number of 10-second holds (up to 10) at the final pressure.

A performance index can be used to document an objective measure. Multiply the pressure increase by the number of times the patient can repeat the 10-second holds—with 100 reflecting the holding of a 10-mmHg increase for 10 repetitions.[25]

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Lower Cervical and Upper Thoracic Extensor Activation and Training

Patient position and procedure: Prone with forehead on the treatment table and arms at the sides. Have the patient lift the forehead off the treatment table, keeping the chin tucked and eyes focused on the table to maintain the neutral spinal position (reinforces the craniocervical flexion motion learned in the supine position). Lifting the head is a small motion (Fig. 16.19).

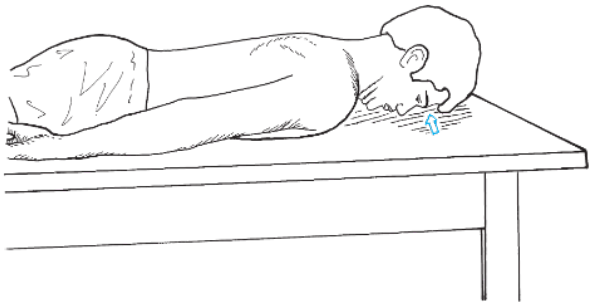


FIGURE 16.19 Axial extension (cervical retraction) exercises.

Progression

Once the patient learns to activate the core musculature and assume the neutral posture in the cervical spine, practice throughout the day is encouraged. Core muscle activation is then coordinated with stabilization training using the global musculature and upper extremity loading. Extremity motions are added and used to stimulate muscular endurance as well as strengthen the stabilizing musculature in the spine. Global stabilization exercises are described in the section Global Muscle Training for Stabilization.

Lumbar Musculature

Three techniques for abdominal muscle activation have been described and used in clinical practice: the drawing-in maneuver; abdominal bracing; and posterior pelvic tilt (Fig. 16.20). Studies are showing that each technique differs in the stabilization activity of the abdominal and multifidus muscles.[42] Research has demonstrated that the drawing-in maneuver is more selective in co-activating the transversus abdominis and multifidus muscles than the abdominal bracing and posterior pelvic tilt techniques.[22,42] The drawing-in maneuver also functions to increase intra-abdominal pressure by inwardly displacing the abdominal wall. Because of this, the drawing-in maneuver is recommended for stabilization training; the other two methods are also described, primarily so the reader can recognize the differences.

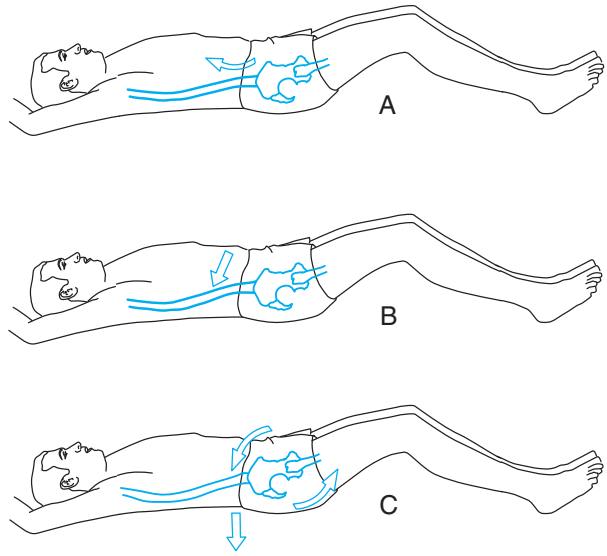


FIGURE 16.20 Three methods to activate the stabilizing musculature in the lumbar spine. (*A*) Drawing-in maneuver where the patient hollows the abdominal region ("draws" the belly button toward the spine). (*B*) Abdominal bracing where setting the abdominal muscles results in flaring laterally around the waist. (*C*) Posterior pelvic tilt where the pelvis is actively tilted posteriorly and the lumbar spine flattens.

Drawing-In Maneuver (Abdominal Hollowing Exercise) for Transverse Abdominis Activation

Patient positions: Training may be easiest in the quadruped position in order to use the effects of gravity on the abdominal wall. Hook-lying (with knees 70° to 90° and feet resting on an exercise mat), prone-lying, or semireclined positions may be used if more comfortable for the patient. It is important to progress training to sitting and standing as soon as possible for functional activities.

Procedure: Teach the patient using demonstration, verbal cues, and tactile facilitation. Explain that the muscle encircles the trunk; and when activated, the waistline draws inward (see Fig. 16.20A). Palpation of the muscle is possible just distal to the anterior superior iliac spine (ASIS) and lateral to the rectus abdominis (Fig. 16.21). When the internal oblique (IO) contracts, a bulge of the muscle is felt; when the TrA contracts, flat tension is felt. The goal is to activate the TrA with minimal or no contraction of the IO. This is a gentle contraction.

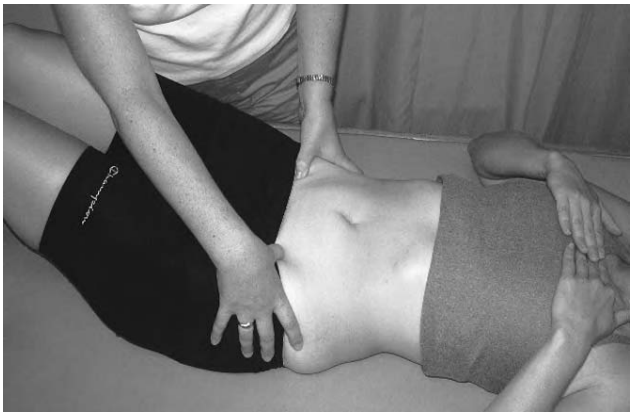


FIGURE 16.21 Palpation of the transversus abdominis muscle just distal to the ASIS and lateral to the rectus abdominis muscle. It feels like a tense sheet (a bulge is the internal oblique) when performing a gentle drawing-in maneuver.

Have the patient assume a neutral spinal position and attempt to maintain it while gently drawing in and hollowing the abdominal muscles.[40] Instruct the patient to breathe in, breath out, then gently draw the belly button in toward the spine to hollow out the abdominal region. When done properly there are no substitute patterns; that is, there is minimal to no movement of the pelvis (posterior pelvic tilting), no flaring or depression of the lower ribs, no inspiration or lifting of the rib cage, no bulging out of the abdominal wall, and no increased pressure through the feet.

If a patient has difficulty activating the TrA, the following two feedback techniques have been shown to assist with learning.[11,26,41,42]

- ***Pressure biofeedback for clinical testing and visual feedback.*** With the patient prone, the Stabilizer™ is placed horizontally under the abdomen (centered under the navel). Inflate the Stabilizer™ to 70 mmHg. Have the patient perform a drawing-in maneuver, as described above. A decrease of 6 to 10 mmHg during the drawing-in maneuver (without substitutions) indicates proper activation of the deep abdominal muscles. The dial on the unit is large and easily read by the patient for immediate feedback.
- ***Biofeedback with surface electrodes.*** Surface electrodes placed over the rectus abdominis and external obliques (near its attachment on the eighth rib) may be used in conjunction with the inflatable cuff. There should be minimal to no activation of these muscles if the drawing-in maneuver is done correctly.

BOX 16.4 Testing and Training Core Muscle Activation (Transversus Abdominis) in the Lumbar Spine

- Patient is prone lying.
- Place the Stabilizer™ pressure biofeedback unit horizontally under the abdomen with the lower edge just below the anterior superior iliac spines (ASIS) (navel at center of unit).
- Inflate to 70 mmHg and instruct the patient to perform the drawing-in maneuver.
- If done properly, the pressure drops 6 to 10 mmHg.
- See if the patient can maintain the pressure drop for up to 10 seconds.
- Muscle endurance (holding or tonic capacity) of the transversus abdominis (TrA) is measured by the number of 10-second holds (up to 10).

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As with the cervical spine, the Stabilizer™ can be used not only to train activation of the TrA but also to measure control for a period of time as well as number of repetitions. The protocol is summarized in Box 16.4.

Abdominal Bracing

In contrast to the drawing-in maneuver, abdominal bracing occurs by setting the abdominals and actively flaring out laterally around the waist (see Fig. 16.20B). There is no head or trunk flexion, no elevation of the lower ribs, no protrusion of the abdomen, and no pressure through the feet. The patient should be able to hold the braced position while breathing in a relaxed manner. This technique has been taught for a number of years as the method to stabilize the spine; and it has been shown to activate the oblique abdominal muscles consistent with their global stabilization function.[42]

Posterior Pelvic Tilt

Pelvic tilt exercises (see Fig. 16.20C) principally activate the rectus abdominis muscle, which is used primarily for dynamic trunk flexion activity. It is not considered a core spinal stabilization muscle; therefore, it is not emphasized in the training for stabilization.[42] It is used mostly to teach awareness of movement of the pelvis and lumbar spine. It is activated when the patient explores his or her lumbar ROM with pelvic tilts to find the neutral position or functional spinal range.

Multifidus Activation and Training

Patient position and procedure: Prone or side-lying. Place your palpating digits (thumbs or index fingers) immediately lateral to the spinous processes of the lumbar spine

(Fig. 16.22). Palpate each spinal level so comparisons in the activation of the multifidus (Mf) muscle can be made between each segment as well as from side to side. Instruct the patient to "swell the muscle" out against your digits. Palpate for consistency of muscle contraction at each level. Facilitation techniques include using the drawing-in maneuver and gently contracting the pelvic floor muscles (as in Kegel's exercises, described in Chapter 23). In the side-lying position, facilitate by gently applying manual resistance to the thorax or pelvis to activate the rotation function of the Mf.

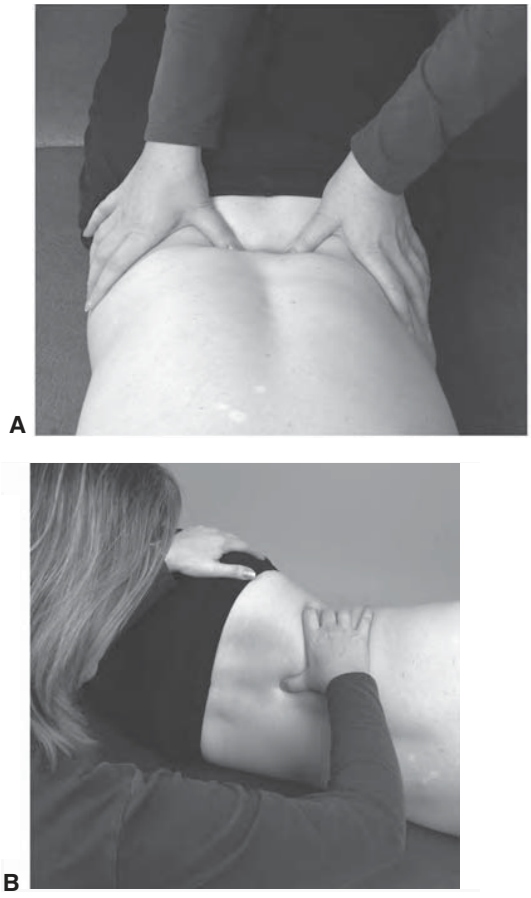


FIGURE 16.22 Palpation of the multifidus muscle lateral to the spinous processes in the lumbar spine, (*A*) bilaterally in the supine position and (*B*) unilaterally in the side-lying position.

The patient may be taught to self-palpate an Mf contraction in the following manner. Sit and rock the pelvis to find the neutral position. With the fingers or thumbs placed along the lumbar spinous processes, lean forward a couple degrees. The Mf is thus activated. Differentiate the Mf contraction from tension in the aponeurosis of the global erector spinae.

Progression

Once the patient learns to activate the core musculature, practice throughout the day is encouraged. Core muscle activation is then coordinated with stabilization training using the global musculature and extremity loading. Extremity motions are added and used to stimulate muscle endurance as well as strengthen the trunk muscles. Global stabilization exercises are described in the next section.

Global Muscle Stabilization Exercises

Even though this section is divided into cervical and lumbar regions, many of the same exercises may be used for impairments in either region because of the functional relationships of the entire axial skeleton.

Stabilization Exercises for the Cervical Region

Stabilization with Progressive Limb Loading

In general, stabilization exercises begin in the recumbent position and progress to sitting, sitting on a large gym ball, standing with the back supported against a wall, and finally standing without support.

- Begin all exercises with gentle craniocervical nods and axial extension to the neutral spinal position. During the early phases of training, if the patient has difficulty maintaining a neutral spinal position, a small towel roll may be placed under the neck for passive support.
- Initially, the only resistance load comes from simple upper extremity movements. When the patient can perform multiple repetitions of the upper extremity motions, resistance is added with handheld weights or elastic resistance.
- The principles of muscle endurance and strengthening described in Chapter 6 are used to challenge the spinal stabilizing musculature.
- Table 16.3 summarizes limb-loading exercises that emphasize the flexor muscles, and Figure 16.23 illustrates the basic exercise progression in the supine position.

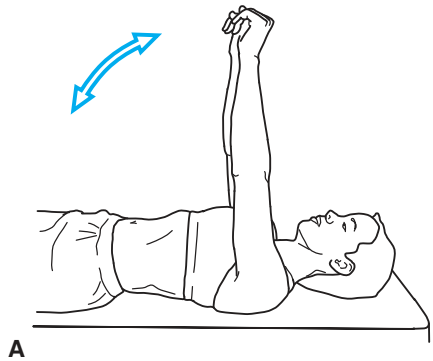


FIGURE 16.23 Limb loading for basic stabilization progression of cervical musculature in the supine position. Maximum protection phase: (*A*) shoulder flexion to 90°;

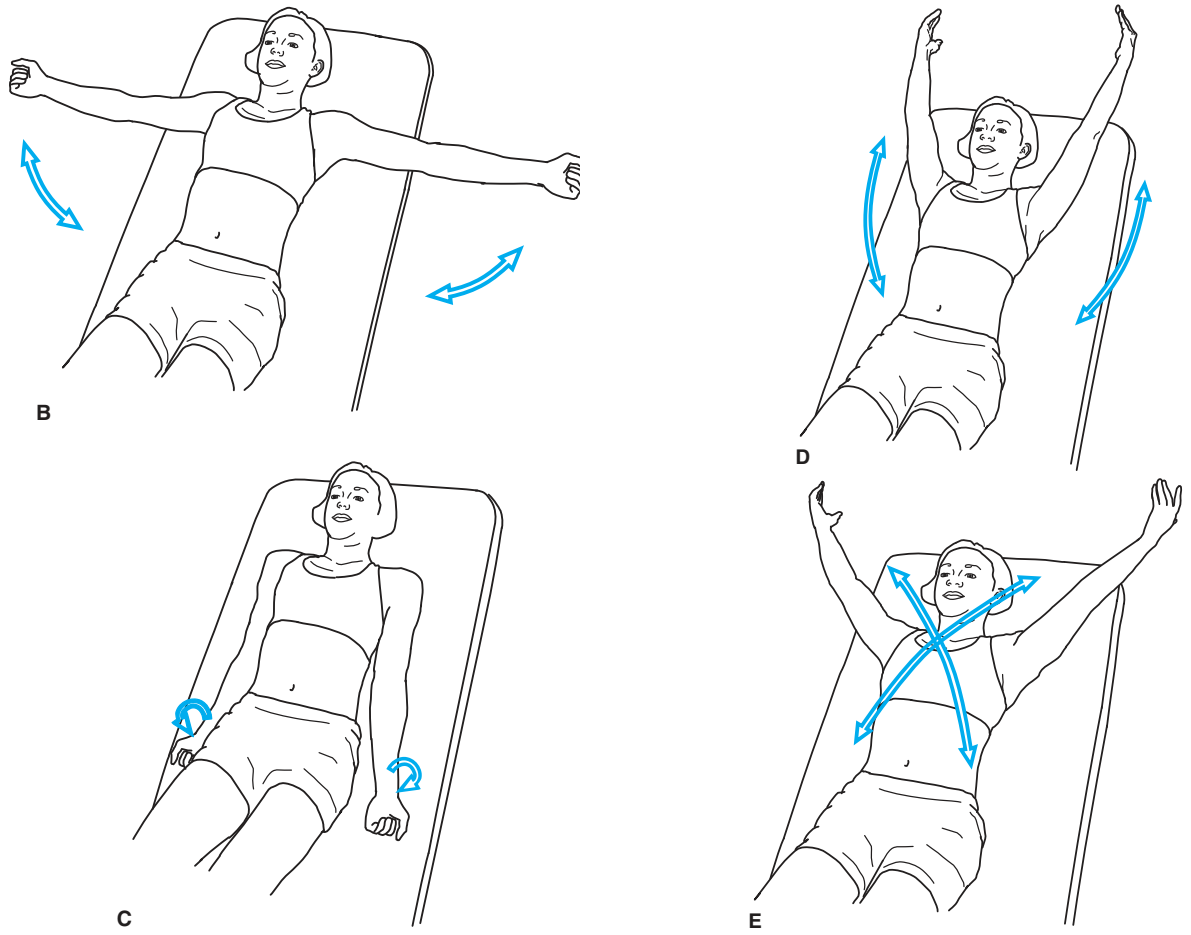


FIGURE 16.23 (continued) (*B*) shoulder abduction to 90°; (*C*) shoulder external rotation arms at the side. Moderate protection phase: (*D*) shoulder flexion and abduction to end range; (*E*) diagonal patterns.

TABLE 16.3 Cervical Stabilization with Progressive Limb Loading—Emphasis on Cervical Flexors

Instructions: Determine amount of support needed and amount of protection. Begin each exercise with axial extension to the neutral spinal position and maintain it while exercising; increase extremity repetitions, then increase resistance before progressing to a new challenge.		*Maximum Support* ⟷ *Minimum Support* *Supine* · *Sitting (sitting on ball for less stability)* · *Standing with wall support* · *Standing with no support*
Core Activation		Gentle craniocervical flexion/axial extension hold 10 seconds × 10 repetitions
Minimum limb loading ↕ **Maximum limb loading**	*Maximum to moderate protection*	Shoulder flexion to 90°
		Shoulder abduction 90°
		Shoulder external rotation with arms at sides
	Moderate to minimum protection	Shoulder flexion to end of range
		Shoulder abduction combined with external rotation to end of range
		Diagonal patterns
	Minimum to no protection	Reaching forward, outward, upward in functional patterns
		Standing, no support: pushing/pulling and lifting activities

TABLE 16.4 Cervical Stabilization with Progressive Limb Loading—Emphasis on Cervical and Thoracic Extensors

		Maximum Support ⟷ *Minimum Support*			
Instructions: Determine amount of support needed and amount of protection. Begin each exercise with axial extension to the neutral spinal position and maintain it while exercising; increase extremity repetitions, then increase resistance before progressing to a new challenge.		Prone forehead on treatment table—lift forehead off table (Fig. 16.19)	Quadruped over padded stool or gym ball—maintain eyes focused on floor	Standing back supported by wall (ball behind head for less stability)	Standing, no support
Core activation—gentle craniocervical flexion/axial extension		Lift forehead off exercise mat; hold 10 seconds × 10 repetitions			
Minimum limb loading ↕ **Maximum limb loading**	*Maximum to moderate protection*	Arms at side: laterally rotate shoulders and adduct scapulae			
		Arms in 90/90 position (abducted and laterally rotated), horizontally abduct shoulders and adduct scapulae			
	Moderate to minimum protection	Elevate shoulder in full flexion			
		Arms abducted to 90° and laterally rotated, elbows extended: horizontally abduct shoulders and adduct scapulae			
		Upper extremity diagonal patterns			
	Minimum to no protection	Standing: reaching forward, outward, upward in functional patterns			
		Standing, no support: pushing/pulling and lifting activities			

Table 16.4 summarizes limb-loading exercises that emphasize the lower cervical/upper thoracic extensor muscles, and Figure 16.24 illustrates a basic exercise progression in the prone position. It is important to note that these exercises do not isolate the flexors or extensors, but that the designation is primarily for emphasis due to the effects of gravity.

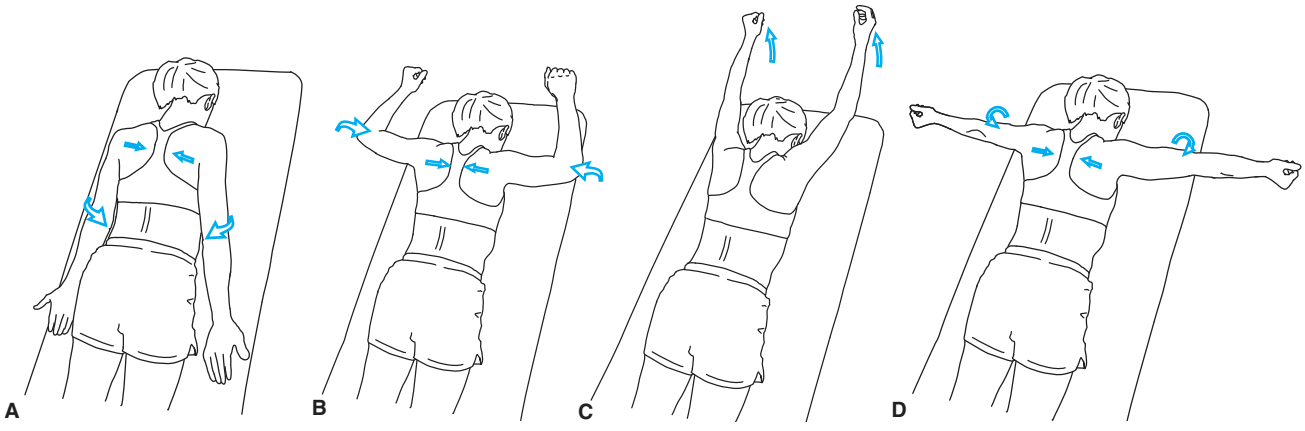


FIGURE 16.24 Limb loading for basic stabilization progression of cervical musculature in prone position. Maximum protection phase: (*A*) arms at side, shoulder lateral rotation, and scapular adduction; (*B*) arms at 90/90, horizontal abduction, and scapular adduction. Moderate protection phase: (*C*) shoulder elevation full range, (*D*) shoulders 90° with lateral rotation and elbow extended, horizontal abduction, and scapular adduction.

Variations and Progressions in the Stabilization Program

Remind the patient to find and maintain the neutral spinal position when doing these exercises.

- ***Extremity loading.*** During the early phases of training, limit shoulder flexion to 90° flexion and abduction. Once the patient can maintain stability and symptoms are not provoked, greater challenges occur with elevating the upper extremity full ROM. Unilateral upper extremity motion requires greater control than bilateral motion because the activity is asymmetrical.
- ***External resistance.*** Tables 16.3 and 16.4 summarize progressions based on position changes. In addition, use of resistance loads (free weights or elastic resistance) adds to the stabilizing challenge. Even though external resistance applied through the extremities has the benefit of increasing strength in the extremity musculature, the primary goal is to increase the stabilizing response of the cervical musculature. Therefore, any loss of the neutral spinal posture or increase in cervical symptoms signals the need to decrease the intensity of the resistance force.
- ***Unstable surfaces.*** Use of a ball while sitting (Fig. 16.25A), when in the quadruped position (Fig. 16.25B), or when standing supporting the ball between the head and the wall (Fig. 16.25C) provides an unstable surface that challenges the muscles to respond to perturbations.



FIGURE 16.25 Unstable surfaces provide increased challenges to the cervical stabilizing musculature, requiring greater control. Examples include performing upper extremity motions, such as diagonal patterns, (*A*) while sitting on a gym ball, (*B*) while quadruped over a gym ball, and (*C*) while pressing a ball against the wall. Use of external resistance is also illustrated.

- ***Muscular endurance and strength.*** Determine the maximum level of resistance tolerated by the cervical-stabilizing musculature and that does not reproduce symptoms. Decrease the intensity and have the patient exercise with multiple repetitions at that level (20 to 30 repetitions or for 1 minute). Resistance can then be added for strengthening (decrease the number of repetitions) at that level before progressing to endurance training at the next level.

Integration of Stabilization Exercises and Posture Training

Good postural alignment of the neck begins with the pelvis and lumbar spine and moves on up to the scapular and thoracic regions. The thorax must be lifted up from the pelvis and scapula retracted in a comfortable position for the cervical spine to assume an efficient position of axial extension (cervical retraction). Therefore, begin with lumbopelvic control if necessary and develop thoracic extension and scapular retraction. While the patient is performing the extremity motions to develop stability, reinforce good scapulohumeral alignment. It is important to remember that strengthening alone does not correct faulty posture and therefore to utilize reinforcement techniques and environmental adaptations that are discussed in Chapter 14.

Progression of Isometric and Dynamic Strengthening in Conjunction with Functional Activities

When the patient demonstrates good cervical stabilization and response to various upper extremity resistance changes, isometric and dynamic exercises are integrated

into the program. These are described in the Isometric and Dynamic Exercise section following this section.

Stabilization Exercises for the Lumbar Region

Stabilization with Progressive Limb Loading

Once the patient learns to activate the core-stabilizing muscles in the lumbar region, explain that prior to each exercise the patient is to find the neutral spinal position, perform the drawing-in maneuver, and then maintain control while applying an exercise load with extremity motions. The drawing-in maneuver develops the pattern of setting the deep abdominal and multifidus muscles in a feedforward pattern and then trains their holding capacity in coordination with the global muscles.[11,26] If the patient cannot control the position, preposition him or her using pillows, supports, or corsets

- To improve the holding capacity of the stabilizing muscles, increase the amount of time the patient does the exercises. It is important that no exercise is continued if the patient cannot maintain the stable position. If the deep abdominals cannot stabilize, substitute patterns in the superficial muscles override the deep muscle activation.
- The Stabilizer™ Pressure Bio-Feedback unit may be used for feedback during this early training (see Box 16.5 for guidelines).
- Table 16.5 summarizes basic limb loading exercises in the supine position that emphasize the abdominal muscles, and Figures 16.26 and 16.27 illustrate the exercise progression.

BOX 16.5 Instructions for use of Stabilizer™ for Stabilization Training with Leg Loading

Patient position: Supine, hook lying.

- Place the three-chamber pressure cell under the lumbar spine horizontally across low back area.
- Position the spine in neutral.
- Inflate the pressure cell to a baseline of 40 mmHg.
- Draw in the abdominal wall without moving the spine or pelvis.
- Pressure should remain at 40 mmHg (±10 mmHg) while performing the lower extremity loading exercises.

Adapted from the instruction manual that accompanies the Stabilizer™ © 2006 Encore Medical, L.P., with permission.

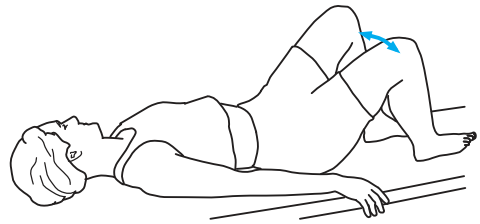


FIGURE 16.26 Bent-leg fall out. Level 2 limb loading for basic stabilization of the abdominal muscles in the supine position. This requires control to prevent pelvic rotation; stability is assisted by the opposite lower extremity while hook-lying.

TABLE 16.5 Basic Lumbar Stabilization with Progressive Limb Loading—Emphasis on Abdominals

Instructions: Patient position hook lying (knees 90°). Place pressure cuff under lumbar spine and inflate to 40 mmHg. Begin each exercise with drawing-in maneuver to activate core muscles. Determine level at which patient can maintain pressure constant (stable pelvis) while performing either A, B, or C limb load activity. For endurance, decrease load and perform repetitive motion for 1 minute or longer. For strength, progress load.		*Progressive Limb Loading* →		
		A. Lift bent leg to 90° hip flexion	B. Slide heel to extend knee	C. Lift straight leg to 45°
Minimum external support	*Level 1: core activation*	Draw in and hold 10 seconds		
Maximum external support	*Level 2:*	Opposite LE on mat; bent leg fall out		
↓	*Level 3: A, B, or C*	Opposite LE is on table		
↓	*Level 4: A, B, or C*	Hold opposite LE @ 90° of hip flexion with UE		
↓	*Level 5: A, B, or C*	Hold opposite LE @ 90° of hip flexion (no UE assistance)		
Minimum external support	*Level 6: A, B, or C*	Bilateral LE movement		

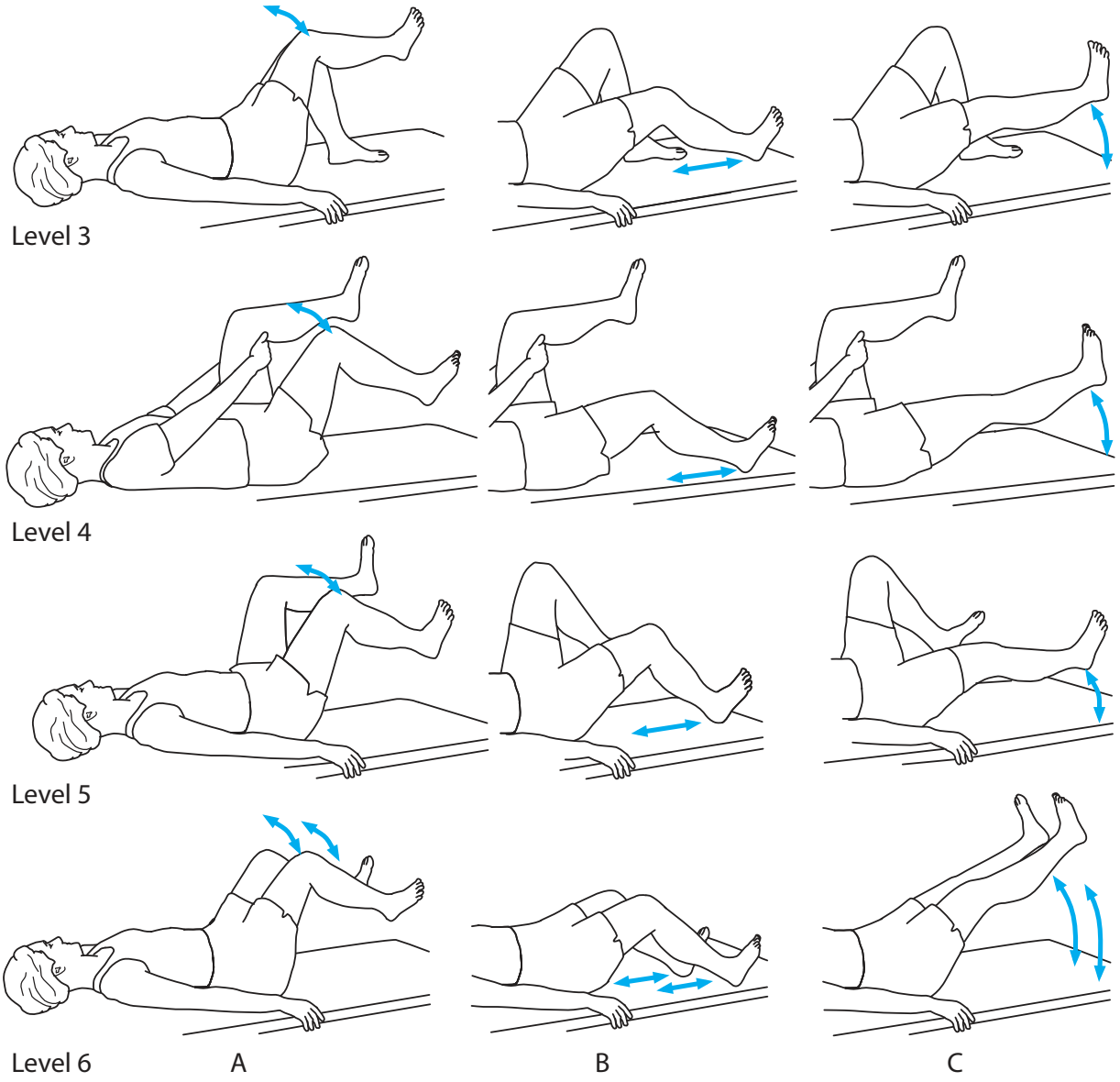


FIGURE 16.27 Limb loading for basic stabilization progression of the abdominal muscles in the supine position, levels 3 to 6. Level 3, stability assisted by opposite extremity in hook-lying position; level 4, stability assisted by patient holding opposite leg at 90°; level 5, stability challenged by patient actively holding opposite leg at 90°; level 6, stability challenged with both lower extremities moving. (*A*) Bent leg lift to 90°. (*B*) Heel slide to extend knee. (*C*) Straight-leg lift to 45°.

Table 16.6 summarizes limb-loading exercises in the quadruped and prone positions that emphasize the extensor muscles, and Figure 16.28 illustrates a basic exercise progression. Performance of extremity loading in the prone position places a greater compressive load on the lumbar spine[4,33] and is not possible if there are hip flexion contractures; therefore, extension exercises are initiated in the quadruped position so the lumbar spine can more easily be positioned in neutral and the patient can learn control. If the patient cannot bear weight on the extremities or maintain balance in the quadruped position, use a padded stool or gym ball for additional support. It is important to maintain the cervical spine in its neutral position during these exercises. The patient should be able to align the head and focus the eyes on the floor. As the exercises progress, there is greater challenge on co-activation of all of the stabilizing musculature.

TABLE 16.6 Basic Lumbar Stabilization with Progressive Limb Loading—Emphasis on Trunk Extensors

		Position	Load
Instructions: Patient position quadruped or prone. Patient assumes neutral spine in lumbar and cervical regions (keeping eyes focused toward floor or exercise mat), performs drawing-in maneuver, and moves extremities. Motions are repeated or alternated from side to side.	Lower intensity	Quadruped position	Flex one upper extremity (UE)
			Extend one lower extremity (LE) by sliding it along the exercise mat
			Extend one (LE) and lift 6–8 inches off exercise mat
			Flex one UE and extend contralateral LE
	Greater intensity and spinal compression	Prone lying position—near end of range of motion, requiring greater control of neutral spine	Extend one LE
			Extend both LE
			Lift head, arms and LE

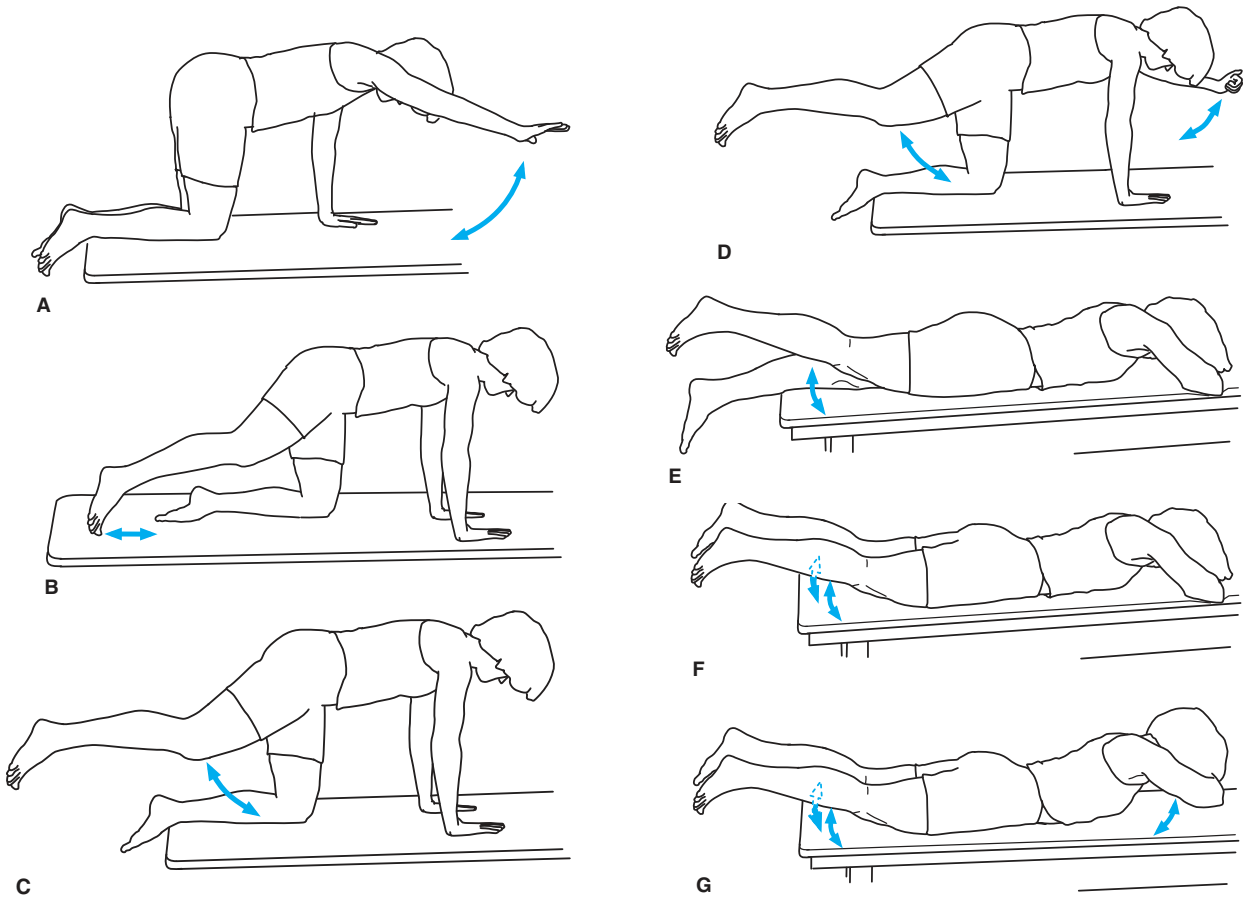


FIGURE 16.28 Limb loading for basic stabilization progression of the lumbar extensors. Begin in the quadruped position and progress the intensity by (*A*) flexing one UE, (*B*) extending one LE with a leg slide, (*C*) extending one LE by lifting it off the mat, (*D*) flexing one UE while extending contralateral LE and then alternate to opposite extremities. Progress to prone: (*E*) extending one LE, (*F*) extending both LE, and (*G*) lifting head, arms, and trunk.

Focus on Evidence

The exercise progressions described in Table 16.5 are adapted from several research studies that investigated the reliability, validity, and sensitivity to change one exercise level with abdominal muscle stabilizing ability using lower-limb loading.[10,11,23] The exercise progressions described in Table 16.6 are adapted from electromyography (EMG) studies that documented extensor activity with limb loading in the quadruped and prone-lying positions.[4,33]

Variations and Progressions in the Stabilization Exercise Program

For all exercises, reinforce the importance of first finding the neutral spine (cervical and lumbar regions), performing the drawing-in maneuver, and then maintaining the neutral spine while superimposing any extremity motions. It is critical to instruct the patient to stop the exercises (or decrease the intensity) as soon as there is a sense that there is loss of control of the stable spinal position. It is important not to progress the patient beyond what he or she is able to control in order to develop the proper muscle response. The emphasis is first on improving the static holding capacity (endurance) of the trunk muscles followed by strengthening. Endurance training of the trunk extensor muscles is related to decreased pain and improving function during the early stages of recovery in patients with subacute low back pain.[8]

- ***Emphasis on muscle endurance.*** Determine a level of exercise that the patient can perform for several repetitions while maintaining a stable spine in the neutral position. Have the patient exercise at that level with the goal of increasing the number of repetitions or the time. Once the patient can perform repetitions for 1 minute, add weights, decrease the repetitions, and emphasize strength. Progress to the next level of difficulty for muscular endurance.
- ***Use of external props.*** Use of the Stabilizer™ pressure biofeedback unit to help the patient with learn control while doing the abdominal stabilization exercises was described earlier (see Box 16.5). For exercises in the quadruped position, if the patient has difficulty controlling the trunk rotation, use a prop such as a dowel rod placed along the spine. Have the patient attempt to keep it balanced while performing the arm and leg exercises (Fig. 16.29). It may be helpful to cue the patient to not shift his or her weight as the extremity is moved—this is difficult to do but is effective in bringing in the stabilizing trunk muscles.
- ***Extremity loading.*** Boxes 16.5 and 16.6 identify a progression of exercises in supine and quadruped/prone positions with extremity loading. Initially have the patient do the motions repetitively; then progress to alternating the extremities or moving all four extremities simultaneously (Fig. 16.30). This requires the stabilizing musculature to adjust to the shifting loads. Motions begin in the sagittal plane and then progress to the transverse plane and diagonal patterns (unilateral and bilateral), where movement away from the midline adds a rotational component and increases the challenge to the stabilizing musculature.
- ***External resistance.*** Use weights, elastic resistance, or pulleys for strengthening. Several suggestions are

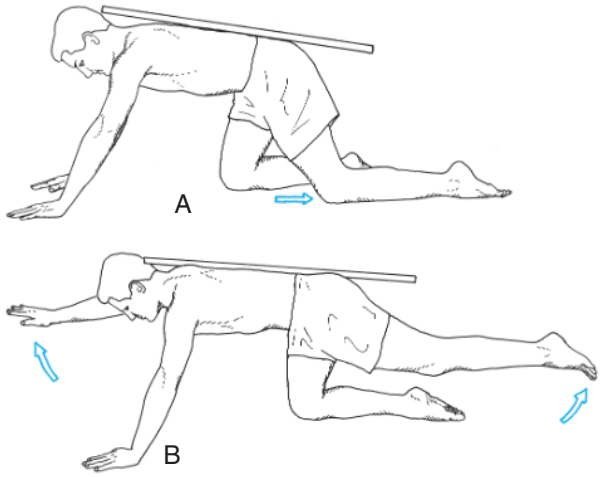


FIGURE 16.29 Balancing a rod on the back while doing quadruped exercises provides reinforcement that the trunk is not twisting. (*A*) Single leg slides. (*B*) Lifting the opposite arm and leg simultaneously, then alternating extremities.

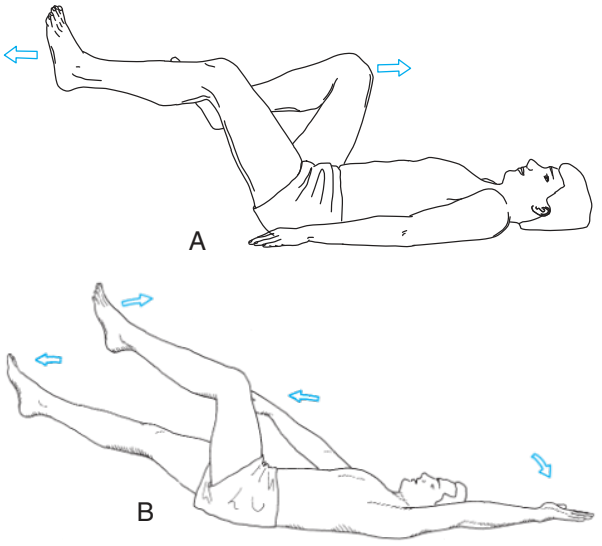


FIGURE 16.30 (*A*) Alternating lower extremity motions with the "modified bicycle" or (*B*) reciprocal and alternating patterns using the upper and lower extremities simultaneously require a strong controlling action in the abdominals.

illustrated in Figures 16.31, 16.32, and 16.33. Even though the extremities benefit from the exercises, the primary purpose is to improve performance in the stabilizing muscles of the trunk; therefore, when signs of fatigue occur, such as poor control of spinal stability (seen as movement of the pelvis or lumbar spine), reduce the intensity or stop the exercise and allow recovery.

- ***Position changes.*** Apply the extremity-loading exercises in the sitting (supported then unsupported), kneeling, and standing positions. Also use modified bridging to challenge the stabilizing function of the trunk musculature. Exercises such as wall slides and partial lunges and bridging with extremity motions use the extremities and trunk during weight bearing and prepare the muscles for functional activities. These exercises are described in the final section under Functional Activities but also serve the purpose of challenging the stabilizing muscles.
- ***Unstable surfaces.*** Use a large gym ball, foam roller, or wobble board to challenge the patient's balance and develop the stabilizing musculature. A variety of positions can be used, such as sitting upright on the ball with the feet on the floor, lying supine with the trunk on the ball and feet on the floor or with the feet on a low mat. If a foam roller is used, the patient is supine, with the roller placed longitudinally along the spine. Examples are shown in Figures 16.34 and 16.35. Secure elastic or pulley resistance in front of the patient at various heights and have him or her do pulling motions (see Fig. 16.33A).

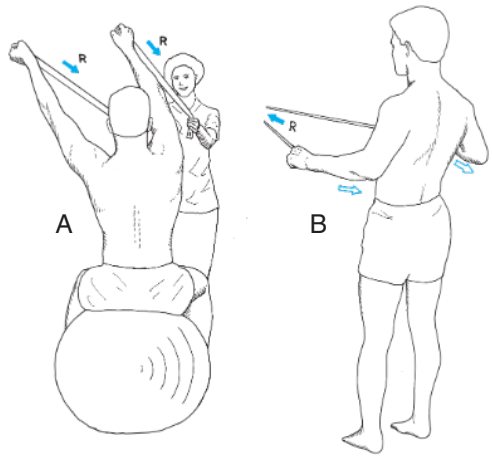


FIGURE 16.33 Using elastic resistance to train and strengthen the back extensor muscles to stabilize in the upright position (*A*) diagonal patterns while sitting on an unstable surface and (*B*) while standing.

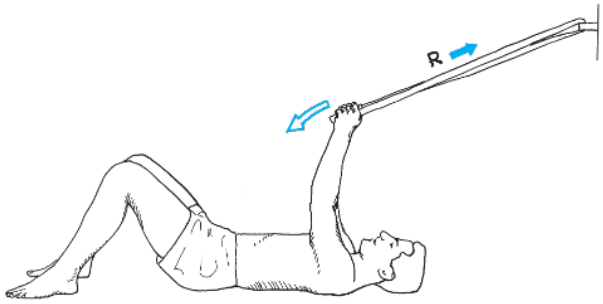


FIGURE 16.31 Developing the stabilizing action of the abdominal muscles by using pull-down activities against a resistive force from pulleys or elastic bands. This exercise can also be done sitting or standing to increase the challenge to the muscles in less stable positions.

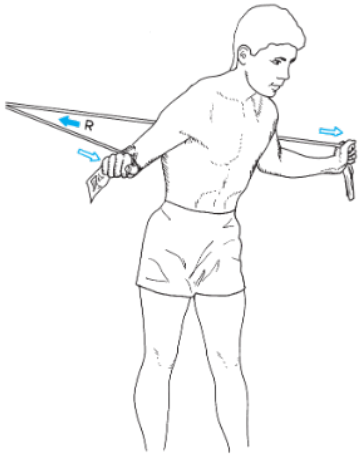


FIGURE 16.32 Using elastic resistance to train and strengthen the abdominal muscles in the upright position. The drawing-in maneuver to set the stabilizing muscles precedes the movement of the arms forward against the resistance.



FIGURE 16.34 Strength, balance, and coordination are required to maintain spinal stabilization while sitting on a gym ball and moving the extremities. This activity is progressed by adding weights to the extremities.

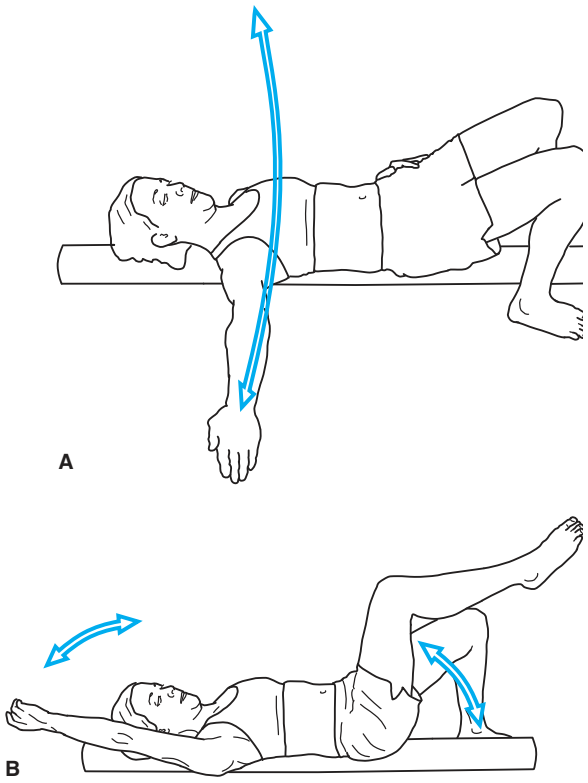


FIGURE 16.35 Activation of the core and global trunk muscles occur to maintain balance on a foam roll while the extremities move in various planes: for example (*A*) shoulder horizontal abduction/adduction and (*B*) ipsilateral hip and shoulder flexion/extension.

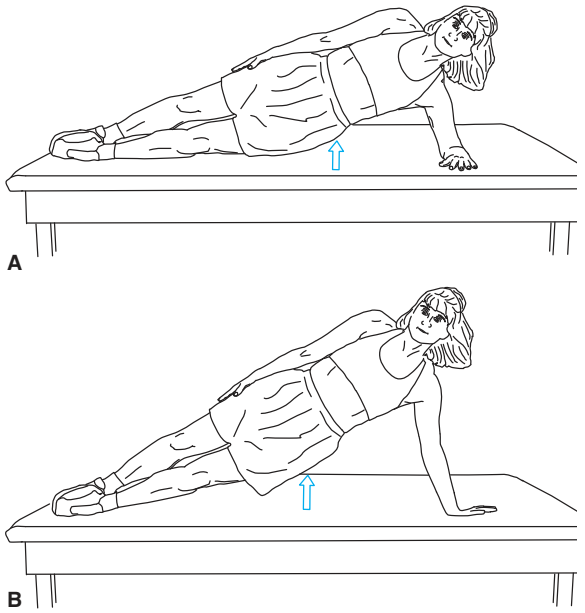


FIGURE 16.36 Quadratus lumborum stabilization training using closed-chain side-propping (*A*) on the elbow and knee and (*B*) on the hand and foot.

Quadratus Lumborum—Stabilization Exercises

The quadratus lumborum has been identified as an important stabilizer of the spine in the frontal and transverse planes.[33] Strongest activation of this muscle occurs with the side propping position. The external obliques are also activated in this position.[33]

Patient position and procedure: Begin side-lying. Have the patient prop up on the elbow and then lift the pelvis off the mat, supporting the lower body with the lateral side of the knee on the downward side. The position can be maintained for an isometric hold or performed intermittently (Fig. 16.36A). Progress by having the patient support the upper body with the hand (with the elbow extended) and lateral aspect of the foot on the downward side (Fig. 16.36B).

Progression to Dynamic Exercises

When the patient has developed control, endurance, and strength in the stabilizing muscles in weight-bearing and non-weight-bearing positions, *dynamic trunk strengthening* exercises are initiated at a low intensity (see following section). The emphasis is on control and safety.

As the patient returns to his or her instrumental activities of daily living (IADLs) and limited work activities, instruct him or her to incorporate the core activation and stabilization techniques into the activities.

Isometric and Dynamic Exercises

Isometric exercises may be considered stabilizing exercises, as there is little or no movement of the spinal segments. They are included in this section with dynamic exercises, however, because of the method of application of the resistive force; that is, the resistive force is applied directly to the axial skeleton rather than through limb loading, as described in the spinal stabilization section. The decision to use the isometric exercises described in this section must be based on the goals of intervention. The exercises may be combined with the stabilization exercises in a home exercise program.

Dynamic exercises with spinal movement are introduced into the patient's exercise program when the patient demonstrates effective core and global stabilization techniques and has developed endurance in the stabilizing musculature. Dynamic exercises should not be a substitute for stabilization exercises. Because of the load imposed on the spine, they may exacerbate the patient's symptoms if introduced prior to effective stabilization and control. They are important in the total rehabilitation of the individual with neck or low back pain, as dynamic muscle endurance and strength is required for many daily activities as well as manual labor and athletic performance.

Exercises for the Cervical Region

Isometric Exercises—Self-Resistance

The intensity of the isometric exercises can be from light to strong, depending on the patient's symptoms and tolerance.

Patient position and procedure: Sitting.

- ***Flexion.*** Have the patient place both hands on the forehead and press the forehead into the palms in a nodding fashion while not moving (Fig. 16.37A).

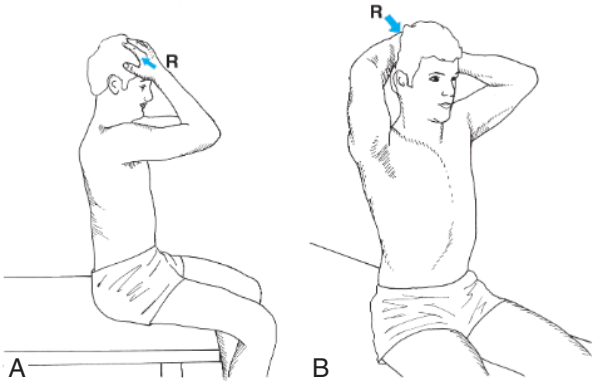


FIGURE 16.37 Self-resistance for isometric (*A*) cervical flexion and (*B*) axial extension.

- ***Side bending.*** Have the patient press one hand against the side of the head and attempt to side-bend, as if trying to bring the ear toward the shoulder but not allowing motion.
- ***Axial extension.*** Have the patient press the back of the head into both hands, which are placed in the back, near the top of the head
- ***Rotation.*** Have the patient press one hand against the region just superior and lateral to the eye and attempt to turn the head to look over the shoulder but not allowing motion.

Isometric Resistance Activities

- *Patient position and procedure:* Standing with a basketball-sized inflatable ball between the forehead and a wall. Have the patient keep the chin tucked and not go into a forward-head posture. The patient maintains the functional position while superimposing arm motions. Progress by adding weights to the arm motions.
- *Patient position and procedure:* Supine with the head over the edge of the mat, the neck maintained in a neutral functional position, and no support to the head. The patient must be able to keep the neck in its safe, functional position to perform this advanced stabilization exercise. He or she holds the position as tolerated. Progress by adding arm motions, then adding weights to the arm motions as tolerated.

Dynamic Cervical Flexion

Often with faulty forward-head postures, the patient substitutes using the sternocleidomastoid (SCM) muscles to lift the head when getting up from the supine position rather than the overstretched, weak, deep cervical flexors. To correct this muscle imbalance, begin training capital flexion as described in the stabilization section (core muscle activation). For home exercise, emphasize "curling" the head and neck, not lifting the head up.

Patient position and procedure: Supine. If the patient cannot tuck the chin and curl the neck to lift the head off the mat, begin with the patient on a slant board or large wedge-shaped bolster under the thorax and head to reduce the effects of gravity (Fig. 16.38). Have the patient practice tucking the chin and curling the head up. Use assistance until the correct pattern is learned. Progress by decreasing the angle of the board or wedge and then adding manual resistance if the patient does not substitute with the SCM.

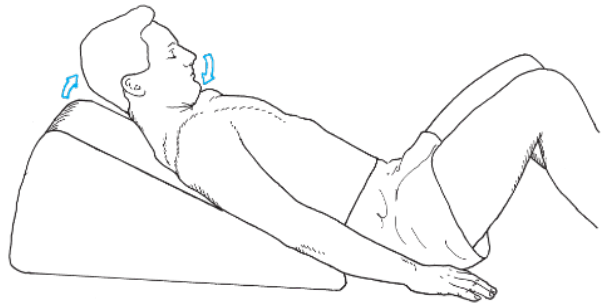


FIGURE 16.38 Training the short cervical flexors while de-emphasizing the sternocleidomastoid for cervical flexion to regain a balance in strength for anterior cervical stabilization.

Manual Resistance—Cervical Muscles

Patient position and procedures: Supine. Stand at the head end of the treatment table, supporting the patient's head for each exercise.

- Place one hand on the patient's head to resist opposite the motion. Do not resist against the mandible lest force be transmitted to the temporomandibular joint. Resistance is given to isolated muscle actions or to general ROMs, whichever best gains muscle balance and function.
- Isometric resistance can be applied with the head in any desired position before applying resistance. Avoid jerking the neck when applying or releasing the resistance by gradually building up the intensity, telling the patient to match your resistance, holding, and then gradually releasing and asking the patient to relax.

Intermediate and Advanced Training

As the patient progresses in the rehabilitation program, greater challenges to the musculature to stabilize and